QUASI-COHERENT MODULES ON A NOETHERIAN GROUPOID SCHEME WITH FLAT QCQS S, T ARE FILTERED COLIMITS OF COHERENT MODULES

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Allowed standard prerequisites (editorial).

- Lemma coherent-lemma-directed-colimit-coherent (canonical tag 0GN6). cited at proof lines 3293-3296. Decomposes F into coherent submodules H_i.
- Lemma lemma-construct-quasi-coherent (07TR) and remark remark-adjunction-map (0GNF). cited at proof lines 3298-3318. The construct functor and the explicit adjunction map.
- Lemma lemma-abelian (077S, its proof) for exactness of restriction. cited at proof lines 3306-3309. Produces the intersection (pullback) diagram.
- Lemma lemma-push-pull (07TT). cited at proof lines 3326-3328. Each F_i is coherent.
- Lemma coherent-lemma-colimit-cohomology (07TB). cited at proof lines 3330-3334. s_* commutes with filtered colimits.

Source reference. Original lemma 01XZ: sources/stacks-chow/tags.txt, source lines 2538–2538. Cited at proof lines 3291-3293 as the standing characterization of coherent modules; load-bearing for the reading of coherence claims.

Source reference. Original lemma 0GN6: sources/stacks-chow/tags.txt, source lines 20781–20781. Cited at proof lines 3293-3296 to decompose F into coherent submodules H_i; load-bearing.

Source reference. Original lemma 07TR: sources/stacks-chow/tags.txt, source lines 9946–9946. Cited at proof lines 3298-3301 for the functor G -> (s_*t^*G, alpha); load-bearing.

Source reference. Original remark 0GNF: sources/stacks-chow/tags.txt, source lines 20790–20790. Cited at proof lines 3301-3303 and 3314-3318 for the description of the adjunction map identifying the pullback diagram; load-bearing.

Source reference. Original lemma 077S: sources/stacks-chow/tags.txt, source lines 9212–9212. Cited at proof lines 3306-3309 (exactness of restriction, via its proof) producing the pullback square; load-bearing.

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Source reference. Original lemma 07TT: sources/stacks-chow/tags.txt, source lines 9948–9948. Cited at proof lines 3326-3328 to conclude F_i is coherent; load-bearing.

Source reference. Original lemma 07TB: sources/stacks-chow/tags.txt, source lines 9931–9931. Cited at proof lines 3330-3334 to commute s_* with the colimit of the H_i; load-bearing for the final identification.

Lemma 0.2. *Let (U, R, s, t, c) be a groupoid scheme over S. Assume that*

(1) *U, R are Noetherian,*
(2) *s, t are flat, quasi-compact, and quasi-separated.*

Then every quasi-coherent module $(\mathcal{F}, \beta)$ on (U, R, s, t, c) is a filtered colimit of coherent modules.

Proof. We will use the characterization of Cohomology of Schemes, Lemma 01XZ of coherent modules on locally Noetherian scheme without further mention. We can write $\mathcal{F} = \operatorname{colim} \mathcal{H}_i$ as the filtered colimit of coherent submodules $\mathcal{H}_i \subset \mathcal{F}$, see Cohomology of Schemes, Lemma 0GN6. Given a quasi-coherent sheaf $\mathcal{H}$ on U we denote $(s_*t^*\mathcal{H}, \alpha)$ the quasi-coherent sheaf on (U, R, s, t, c) of Lemma 07TR. Consider the adjunction map $(\mathcal{F}, \beta) \to (s_*t^*\mathcal{F}, \alpha)$ in $QCoh(U, R, s, t, c)$, see Remark 0GNF. Set

$$(\mathcal{F}_i, \beta_i) = (\mathcal{F}, \beta) \times_{(s_*t^*\mathcal{F}, \alpha)} (s_*t^*\mathcal{H}_i, \alpha)$$

in $QCoh(U, R, s, t, c)$. Since restriction to U is an exact functor on $QCoh(U, R, s, t, c)$ by the proof of Lemma 077S we obtain a pullback diagram

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & s_*t^*\mathcal{F} \\ \uparrow & & \uparrow \\ \mathcal{F}_i & \longrightarrow & s_*t^*\mathcal{H}_i \end{array}$$

in other words $\mathcal{F}_i = \mathcal{F} \cap s_*t^*\mathcal{H}_i$. By the description of the adjunction map in Remark 0GNF this diagram is isomorphic to the diagram

$$\begin{array}{ccc} \mathcal{F} & \longrightarrow & s_*s^*\mathcal{F} \\ \uparrow & & \uparrow \\ \mathcal{F}_i & \longrightarrow & s_*t^*\mathcal{H}_i \end{array}$$

where the right vertical arrow is the result of applying s_* to the map

$$t^*\mathcal{H}_i \to t^*\mathcal{F} \xrightarrow{\beta} s^*\mathcal{F}$$

This arrow is injective as t is a flat morphism. It follows that $\mathcal{F}_i$ is coherent by Lemma 07TT. Finally, because s is quasi-compact and quasi-separated we see that s_* commutes with colimits (see Cohomology of Schemes, Lemma 07TB). Hence $s_*t^*\mathcal{F} = \operatorname{colim} s_*t^*\mathcal{H}_i$ and hence $(\mathcal{F}, \beta) = \operatorname{colim}(\mathcal{F}_i, \beta_i)$ as desired. □